TYPESAFE JEV SYSTEM ONE · TECHNICAL ARCHITECTURE

0.13s Latency

One-Pager Field Guide

Why Jev? The Missing Mathematical Verifier

Why vector databases search for vibes, while TypeSafe Jev mathematically guarantees truth.

PINNED MODEL
`jev-1.14.0`

The Vector Database Trap ("Vibe Search")

Cosine Fail

Vector databases match *dense semantic similarity*. But similarity is blind to truth:

```
Sentence A: "Item delivered 45 days ago"
Sentence B: "Item delivered 15 days ago"
➔ Cosine Similarity: 0.94 (Matches!) But violates the 30-day policy!
```

✕ Embeddings measure phrasing, not truth invariants.

The Jev System One Solution ("Invariant Gate")

Calibrated $p \in [0, 1]$

Jev scores **atomic propositions (Nouls)** directly over state x:

```
Proposition: "Avoid return for item delivered >30 days?"
➔ Jev Noul Probability: p = 0.02 (Violation caught!)
➔ Spikes Lyapunov Energy V(x) ↑ · Descent Gate triggers rollback!
```

✓ Direct scalar verification independent of the LLM generator.

JEV'S 3 CORE PRIMITIVES IN 60 SECONDS

1 The State Tuple (x)

Structured state $x = (u, y, c)$ bundling the user's ground-truth ticket, the candidate reply, and store policy text.

2 Atomic Nouls (p_j)

15 independent policy propositions evaluated concurrently. Each yields calibrated probability $p_j \in [0.0, 1.0]$.

3 Lyapunov Function (V)

$V(x) = \sum w_j(1 - p_j)$ maps multifaceted policy compliance to an intuitive potential energy well.

The Lyapunov Parabola ("Rolling Downhill")

Slide 3

Height V represents rule violations. The Descent Gate ($\delta = 0.0314$) acts as a **one-way ratchet** on the slope:

x_0 (High $V = 0.2887$) 🔴 Rolls Downhill ➔ x_2 (Safe $V \le 0.08$) 🟢

`Descent Rule: Accept only if V(x_{t+1}) ≤ V(x_t) - δ`

✓ Regressions dropped from 61.7% to 5.0% in empirical testing!

Concentric Confidence Rings

Slide 1

Visualizes the agent converging inward toward Ground Truth:

- 🔴 **Outer: Low ($V > 0.25$)** — Breach / Escalate
- 🟡 **Mid: Medium (0.12-0.25)** — Partially Repaired
- 🔵 **Inner: High (0.08-0.12)** — Near Target Bound
- 🟢 **Bullseye: Truth ($V \le 0.08$)** — Safe to Deliver!

DOES JEV USE A VECTOR DB OR GRAPH DB? (COMPARISON TABLE)

DIMENSION	VECTOR DATABASE (PINECONE)	GRAPH DATABASE (NEO4J)	TYPESAFE JEV SYSTEM ONE
Primary Mechanism	Dense Embedding Search (kNN)	Entity & Triplet Graph Traversal	**Neural-Symbolic Verifier**
Mathematical Output	Cosine distance (similarity)	Matched nodes & edges	**Calibrated probabilities $p_j \in [0, 1]$**
Critical Failure Mode	Fails on negation & exact dates	Brittle ontologies; high maintenance	**Requires noise margin $\delta > 2\hat{\eta}$**
Enables Descent Gate?	❌ No (No energy function)	❌ No (Discrete queries only)	✅ **Yes: $V(x) = \sum w_j(1 - p_j)$**

🔑 **The Core Engineering Law: Don't Let Your Generator Grade Itself**

LLMs are proposers. Jev is the independent System One verifier that keeps them rolling downhill.

Noise $\hat{\eta}$ = **0.0126**
Margin δ = **0.0314**